

## Right-angled triangles



1

a AB

b BC

2

a AC

b BC

c BC

d AC

e  $\angle ACB$

3

a i  $\frac{4}{3}$  ii  $\frac{4}{5}$  iii  $\frac{3}{5}$

b i  $\frac{15}{8}$  ii  $\frac{15}{17}$  iii  $\frac{8}{17}$

c i  $\frac{21}{20}$  ii  $\frac{21}{29}$  iii  $\frac{20}{29}$

d i  $\frac{12}{5}$  ii  $\frac{12}{13}$  iii  $\frac{5}{13}$

e i  $\frac{24}{7}$  ii  $\frac{24}{25}$  iii  $\frac{7}{25}$

f i  $\frac{28}{45}$  ii  $\frac{28}{53}$  iii  $\frac{45}{53}$

## Trigonometric ratios

4

a i  $\tan \theta = \frac{AC}{BC}$  ii  $\sin \alpha = \frac{BC}{AB}$

b i  $\sin \theta = \frac{AC}{AB}$  ii  $\tan \alpha = \frac{BC}{AC}$

c i  $\cos \theta = \frac{BC}{AB}$  ii  $\cos \alpha = \frac{AC}{AB}$

d i  $\tan \theta = \frac{8}{15}$  ii  $\sin \alpha = \frac{15}{17}$

e i  $\cos \theta = \frac{15}{17}$  ii  $\tan \alpha = \frac{15}{8}$

f i  $\sin \theta = \frac{24}{25}$  ii  $\cos \alpha = \frac{24}{25}$

5

a  $x = 10$

b  $\frac{4}{5}$

c  $\frac{3}{5}$

6

a  $x = 4$

b  $\frac{4}{3}$



## Unknown sides

7  $b = 8$

8  $d = 3084$

9  $b = 19.8$

10  
a  $f = 4.08$

b  $f = 6.12$

c  $g = 9.46$

d  $a = 11.59$

## Unknown angles

11  $\frac{\sqrt{15}}{7}$

12  
a  $x = 67^\circ$

b  $x = 74^\circ$

13  $A = 66.4^\circ$

14  $\tan \theta = \frac{24}{7}$

## Applications

15  
a  $h = 21$   
b  $f = 22$   
c  $h = 98, x = 28$   
d  $x = 34$   
e  $x = 12$

16  $d = 1.48$

17  $1.96 \text{ m}$

18  $y = 58.02$

19

**a**  $x = 21.85$



**b**  $h = 23.95 \text{ m}$

**20**  $\theta = 9.88^\circ$

**21**  $y = 54.90^\circ$

**22**  $\tan \theta = \frac{4}{3}$

**23**  $h = 4.4 \text{ m}$